

DEKALB-PEACHTREE, GA, USA

WMO: 722196

Lat: 33.875N

Lon: 84.302W

Elev: 305

StdP: 97.71

Time Zone: -5.00 (NAE)

Period: 98-19

WBAN: 53863

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a) 1	(b) -6.0	(c) -3.5	(d) -15.1	(e) 1.0	(f) -1.7	(g) -12.9	(h) 1.3	(i) -0.1	(j) 10.1	(k) 1.9	(l) 9.1	(m) 3.5	(n) 3.7	(o) 310	(p) 0.378

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	10.0	34.5	23.1	33.3	23.0	32.4	22.7	25.0	31.1	24.4	30.3	24.0	29.5	3.2	310

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.1	18.5	26.3	22.8	18.2	26.2	22.4	17.8	26.0	77.1	31.2	75.2	30.7	73.3	29.7	29.3

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
1%	2.5%	5%	Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
			Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
8.3	7.4	6.3	-9.3	36.1	2.5	1.9	-11.1	37.5	-12.6	38.6	-14.0	39.7	-15.8	41.1		
			-10.6	26.0	2.3	0.9	-12.2	26.6	-13.5	27.2	-14.8	27.7	-16.4	28.3		
			DB													
			WB													

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	16.9	6.1	8.3	12.1	16.7	21.3	25.2	26.6	26.4	23.3	17.3	11.5	7.7
	DBStd	8.29	5.50	4.85	4.78	3.88	3.31	2.27	1.92	2.05	2.94	4.15	4.34	4.80
	HDD10.0	408	147	83	34	3	0	0	0	0	0	3	33	106
	HDD18.3	1559	381	281	197	75	14	0	0	0	3	69	208	332
	CDD10.0	2929	24	36	100	203	349	456	516	507	398	228	78	33
	CDD18.3	1038	0	1	6	26	105	206	257	249	151	36	3	1
	CDH23.3	9192	1	2	48	246	886	1881	2435	2281	1152	244	15	1
	CDH26.7	3644	0	0	4	41	264	795	1072	1004	418	46	1	0
Wind	WSAvg	2.4	3.1	3.0	3.1	2.8	2.4	2.2	2.1	1.8	1.8	2.1	2.3	2.7
Precipitation	PrecAvg	1302	112	117	135	93	100	109	121	121	102	83	109	104
	PrecMax	1865	184	197	240	191	285	252	330	261	319	245	221	253
	PrecMin	829	58	45	48	19	32	24	35	47	6	5	26	27
	PrecStd	257	33	42	53	40	53	57	68	47	76	58	51	51
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	22.0	23.2	27.1	29.7	32.6	35.3	36.0	36.3	34.0	30.1	25.8	22.6
		MCWB	15.5	16.5	16.7	18.2	20.9	22.4	23.4	23.4	21.7	21.0	17.2	17.0
	2%	DB	18.9	20.4	24.6	27.8	31.1	33.8	34.2	34.5	32.4	27.7	22.8	20.3
		MCWB	14.2	14.5	15.9	18.2	20.6	22.7	23.4	23.4	21.6	19.2	16.1	16.6
	5%	DB	17.1	18.6	22.6	26.3	29.5	32.5	33.0	32.9	31.0	26.2	21.1	17.9
		MCWB	13.5	13.5	14.9	17.0	20.0	22.5	23.5	23.2	21.7	18.2	15.4	15.0
	10%	DB	14.9	16.9	20.6	24.4	27.9	31.1	32.0	31.7	29.1	24.1	19.0	15.9
		MCWB	11.7	12.2	13.9	16.3	19.7	22.1	23.3	23.0	21.2	17.4	14.4	12.7
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	17.3	17.9	18.4	20.8	23.2	25.1	25.7	25.5	24.1	22.9	20.6	19.5
		MCDB	19.1	20.6	22.9	26.1	28.1	31.5	31.7	32.0	29.1	27.1	21.9	20.4
	2%	WB	15.8	16.6	17.5	19.8	22.2	24.4	25.0	24.9	23.5	21.6	18.5	17.4
		MCDB	18.0	19.2	22.1	24.8	27.7	30.5	31.3	31.0	28.3	25.3	20.5	18.9
	5%	WB	14.2	15.1	16.5	18.9	21.5	23.8	24.5	24.3	22.9	20.6	17.1	15.7
		MCDB	16.4	17.5	20.7	23.6	27.0	29.6	30.4	30.0	27.5	23.6	19.4	17.7
	10%	WB	12.3	13.3	15.3	17.9	20.8	23.1	24.0	23.9	22.4	19.4	15.6	13.3
		MCDB	14.2	15.8	19.3	22.3	25.9	28.6	29.5	29.2	26.6	22.7	18.4	15.0
Mean Daily Temperature Range	5% DB	MDBR	10.3	10.6	11.5	11.8	11.2	10.5	10.0	10.0	10.2	11.4	11.4	9.7
		MCDBR	12.2	12.5	14.0	14.1	12.8	12.4	11.4	11.8	12.4	12.9	13.4	11.6
		MCWBR	8.2	7.8	6.8	5.9	4.4	3.8	3.4	3.4	4.1	5.8	7.3	8.1
	5% WB	MCDBR	10.0	10.1	11.0	10.9	10.3	10.2	10.0	9.9	9.4	9.3	9.7	9.8
		MCWBR	8.0	7.8	6.5	5.7	4.3	3.9	3.4	3.3	3.8	5.4	7.2	8.0
Clear-Sky Solar Irradiance	taub	0.310	0.314	0.347	0.389	0.444	0.474	0.515	0.515	0.418	0.364	0.334	0.310	
	taud	2.539	2.523	2.453	2.320	2.204	2.161	2.062	2.050	2.308	2.455	2.483	2.556	
	Ebn at Noon	908	940	929	899	848	819	783	775	844	870	868	887	
	Edh at Noon	82	92	107	128	145	151	166	165	122	97	86	76	
All-Sky Solar Radiation	RadAvg	2.67	3.23	4.33	5.46	6.04	6.18	5.91	5.46	4.72	3.98	2.97	2.26	
	RadStd	0.16	0.40	0.35	0.36	0.49	0.49	0.43	0.31	0.46	0.49	0.31	0.24	

Historical Trends

	DBAvg	Heating		Cooling			Degree-Days			
		99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
Station Only	(d) N/A	(e) N/A	(f) N/A	(g) N/A	(h) N/A	(i) N/A	(j) N/A	(k) N/A	(l) N/A	(m) N/A
Regional (40 neighbors)	+0.46	N/A	N/A	N/A	N/A	N/A	N/A	N/A	+147	+103

Nomenclature: See separate page